

Semantic Data Charter (SDC4): Architectural & Commercial Briefing

A Strategic Appraisal for Enterprise Architects and Investors. Date: 29 January 2026.

Executive Summary

SDC4 is a 'self-describing interoperability architecture' that explicitly decouples data structure (XSD) from semantic meaning (RDF/OWL). Currently transitioning from a 25-year R&D phase (Founder-led, academic grants) to a commercial SaaS model via Axis SDC, Inc.

Key Differentiator:

Unlike flexible schemas that degrade over time, SDC enforces permanence through a strict "restriction-only" conformance model. The `sd4.xsd` artefact is the immutable source of truth.

Due Diligence Snapshot



Technical Maturity

High. Rigorous XSD 1.1 & W3C alignment.



Adoption Level

Emerging. 11 Repos, ~80 Forks, no public case studies.



Governance Posture

Strict. Schema-first, normative validation.

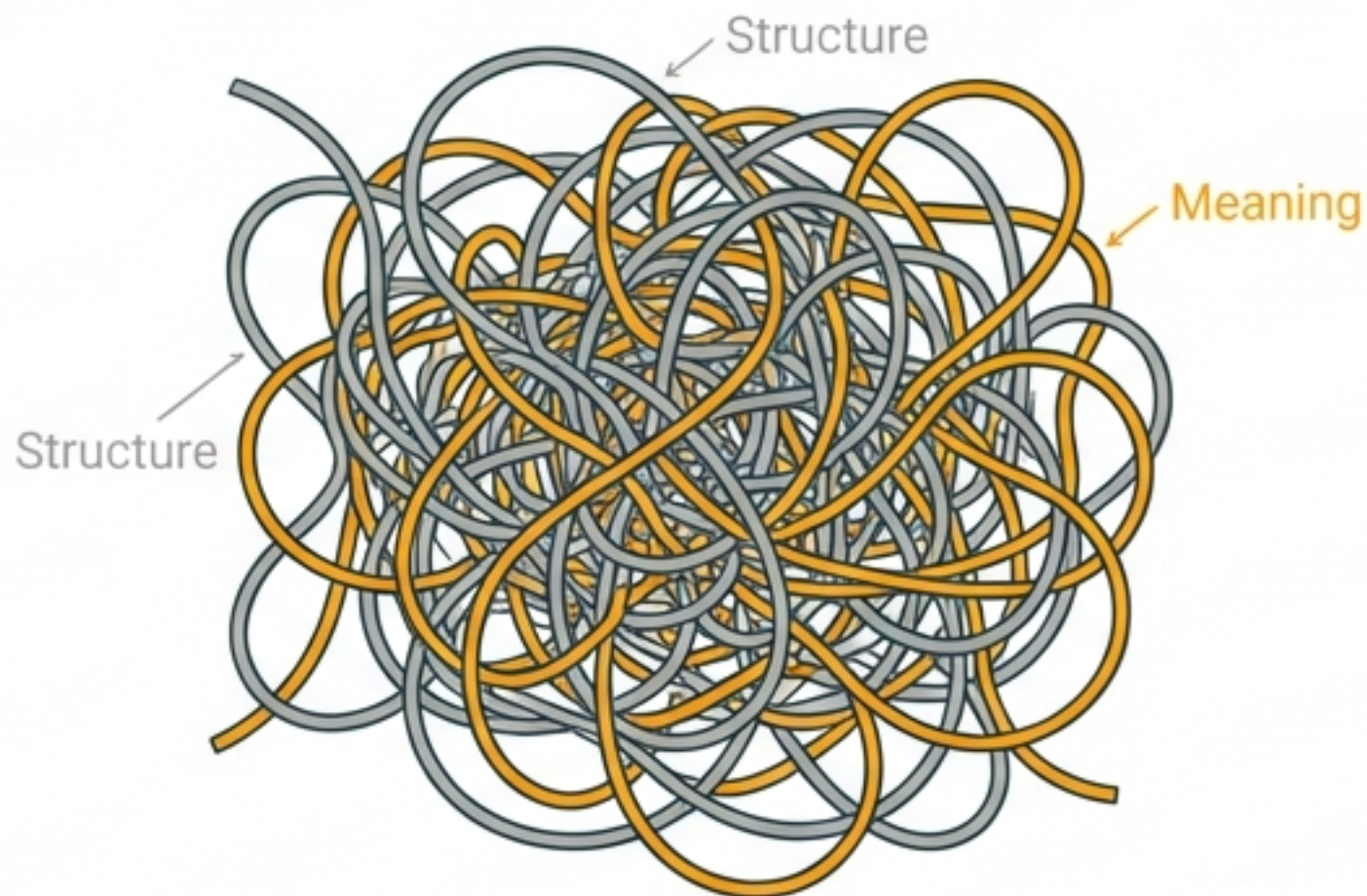


Commercial Status

Bootstrapped. Product-led SaaS (Axis SDC).

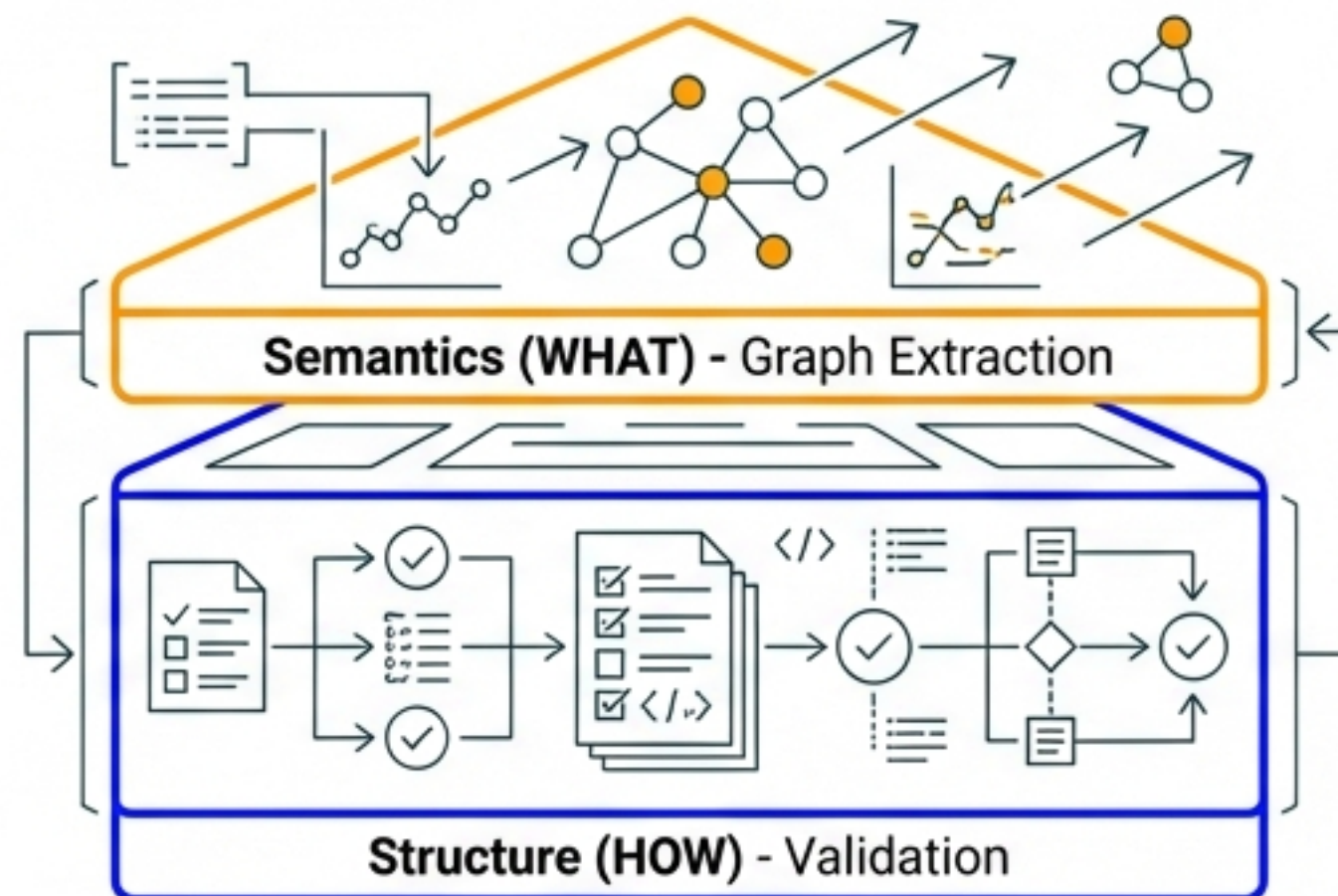
Solving the 'Conflation' Problem in Data Architecture

The Conflation Trap (Current State)



Tightly interwoven structure/meaning increases migration burdens.

The SDC Decoupling (Future State)



Validate structure first, then enrich meaning.

SDC Principles



Enforce Governance

Structure must be validated first.



Embed Meaning

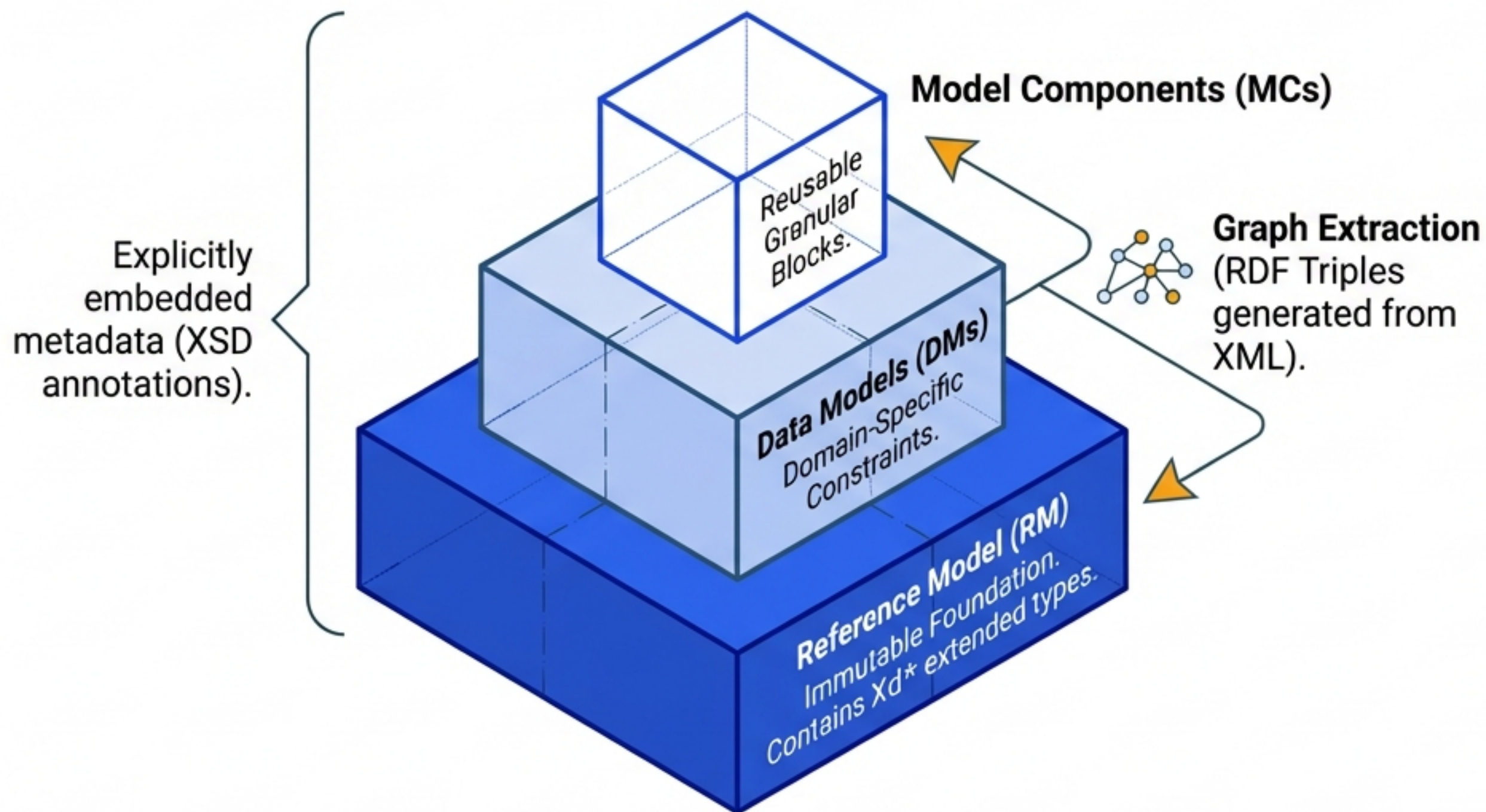
Semantics carried via annotations, not structural hacking.



Mandate Quality

Invalid data is quarantined using ISO 21090 patterns.

The Compositional Hierarchy: Reference Models to Components



Technical Specifications

- **Reference Model Status:** Proprietary Standard (Axis-SDC, Inc.)
- **Core Data Types:** XdStringType, XdTemporalType, XdLinkType
- **Semantics:** Separate Semantic Layer allows Knowledge Graph generation without altering schema.

Governance and Conformance: The Normative Source of Truth

The Golden Rule

sdc4.xsd is the **absolute authority**. If documentation conflicts with the schema file, the schema file is correct.



PERMITTED: xsd:restriction

Adopters must create models by *constraining* the Reference Model.



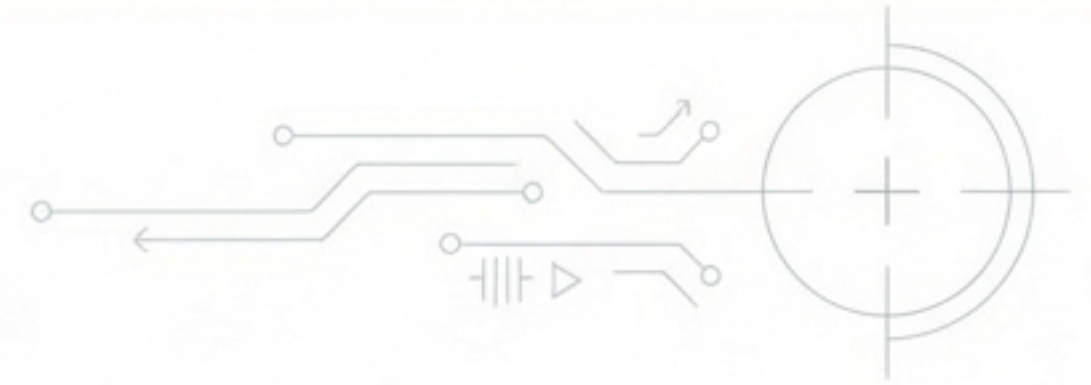
PROHIBITED: xsd:extension

Adopters may *not* add arbitrary elements or attributes to SDC types.

Architectural Implication

Ensures stability. A system built on SDC4 today is guaranteed to validate against the core **Reference** Model in **20 years**. Change control is centralised around the normative schema artefact, requiring a formal RFC process.

The Standards Stack: Leveraging W3C and ISO



Stack Diagram

Data Quality & Formats

- ISO 21090 (Null Flavours)
- ISO 8601 (Temporal)
- RFC 3066/4646 (Language Tags)

Semantic Layer

- RDF 1.1
- OWL 2
- SPARQL 1.1
- SHACL 1.0

Structural Layer

- XSD 1.1 (Advanced Validation Assertions)

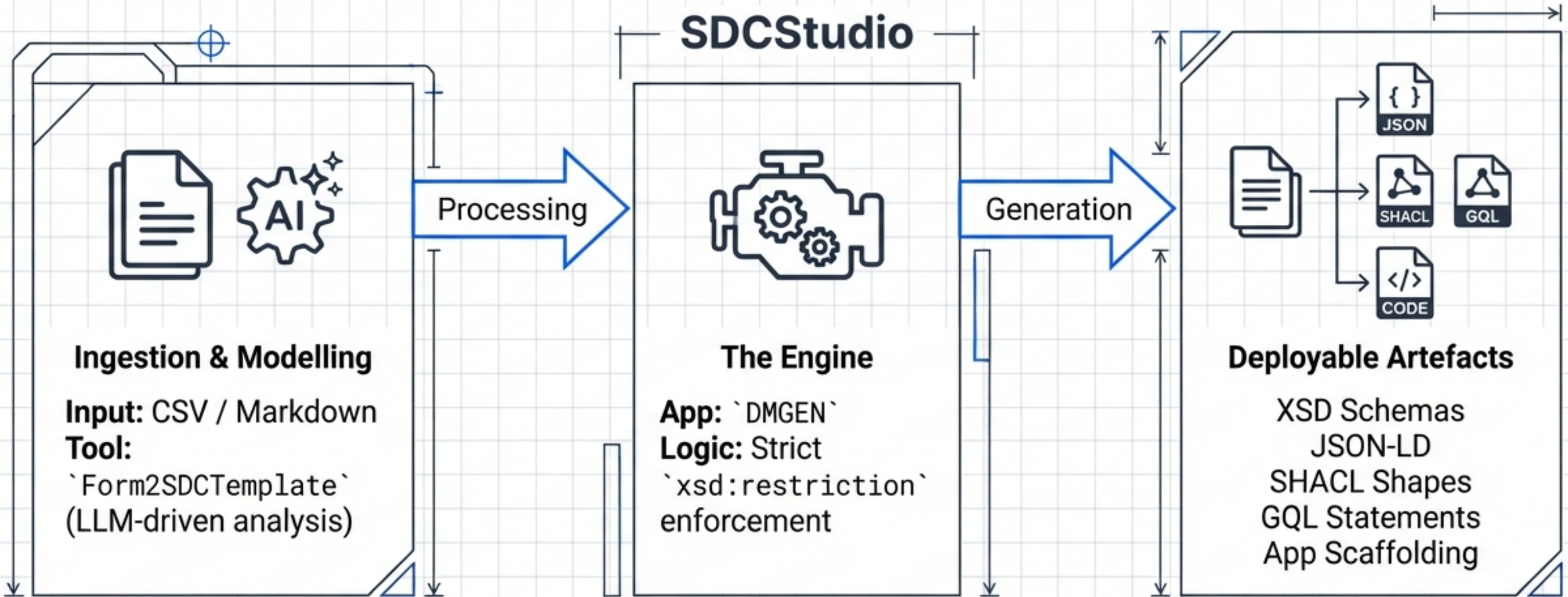
Data Quality Pattern

The '**Quarantine-and-Tag**' Pattern.

Instead of generic nulls, SDC uses ISO 21090 to explicitly tag *why* data is missing.

Not Asked
Masked
Temporarily
Unavailable

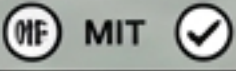
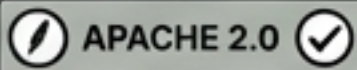
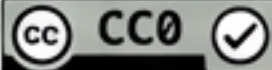
The Operational Pipeline: From Ingestion to Artefact Generation



Infrastructure: AI-Powered (Google Vertex AI Gemini), hosted on Google Cloud Run.

Intellectual Property and Licensing Map

Open Assets: The Code

- **Repositories** (`sdcvalidator`):  MIT
- **Templates** (`ProvGov`):  APACHE 2.0
- **Community Files:** CC0 1.0  CC0

→ Apache 2.0 grants patent rights.

Protected Assets: The Brand

-  **Trademark:** "Semantic Data Charter™" (Owned by Axis SDC, Inc.)
-  **Standard Status:** `sdc4.xsd` is a "Proprietary Standard"

→ Brand identity is controlled. Trademark usage is not automatically granted.

Due Diligence Note:

While the code is open, the 'Proprietary Standard' label suggests stewardship remains with Axis SDC, not a neutral body like OASIS.

Commercial Model: Axius SDC and SDCStudio

Entity

Axius SDC, Inc.

JetBrains Mono

Structure

Founder-owned, Bootstrapped

JetBrains Mono

Product

SDCStudio (Cloud-based SaaS)

JetBrains Mono

Strategy Analysis

The 'Developer Only' Caveat

Inter Tight
Documentation allows local Docker installation but explicitly frames it as "for developers only" and not for production.



Commercial Strategy

Inter Tight
The Charter is the spec; SDCStudio is the adoption accelerator. Revenue model is based on paying for the *convenience* of the managed platform.



Basic

Inter Tight

\$29.99/mo

JetBrains Mono

Single User.
Basic Access.
Limited Projects.

Team

Inter Tight

\$129.99/mo

JetBrains Mono

Up to 5 Users.
Collaboration Tools.
Enhanced Support.

Pro

Inter Tight

\$799/mo

JetBrains Mono

Unlimited Users.
Advanced API Access.
Priority Support.
Custom Integrations.

Enterprise

Inter Tight

Custom

JetBrains Mono

Dedicated Instance.
SLA. White Labeling.
Volume Discounts.
Strategic Partnership.

SEALE

ACCESS LAYER

REVENUE STREAM

Funding Structure and Institutional History

100% Founder-owned. No outside investors. No board pressure.

Source: Axius SDC "Our Story".

R&D Investment

\$8M+ (Claimed)

Bootstrapped R&D over 25+ years.
Self-reported; no breakdown available.



Grant History

Academic Grants (2014-2018)

- UERJ (Brazil) - MLHIM to S3Model Transition
- FAPERJ (Brazil) - Research Grant

Specific grant amounts undisclosed.

⚠ Risk Profile

Risk Profile

The absence of VC funding suggests long-term independence but places reliance on immediate SaaS revenue.

Market Maturity and Adoption Signals



Hard Metrics (Jan 2026)

11

GitHub Repositories

Reality Check

~80

Forks on "sdcvalidator"

Reality Check

0

Public Case Studies
(Website states "Coming Soon")

Adoption Accelerators

NIEM Foundation



NIH Common Data Elements



FHIR Clinical



3,800+ components

500+ CDEs

150+ resources

⚠ Risk Profile

Risk Profile

Analysis: High engineering readiness for government/healthcare, but community is currently in 'Spectator' phase.

Ecosystem Partnerships and Development Roadmap

Partnerships

Infrastructure



Google for Startups Cloud Program

Potential credits up to \$200k/\$350k
(Undisclosed award).

Commercial



Graphwise

Listed as OEM Partner.

Tooling Roadmap



Python Validator (`sdcvalidator`)

Production Ready



TypeScript Validator (`sdcvalidatorJS`)

In Development



Rust Validator (`sdcvalidatorRust`)

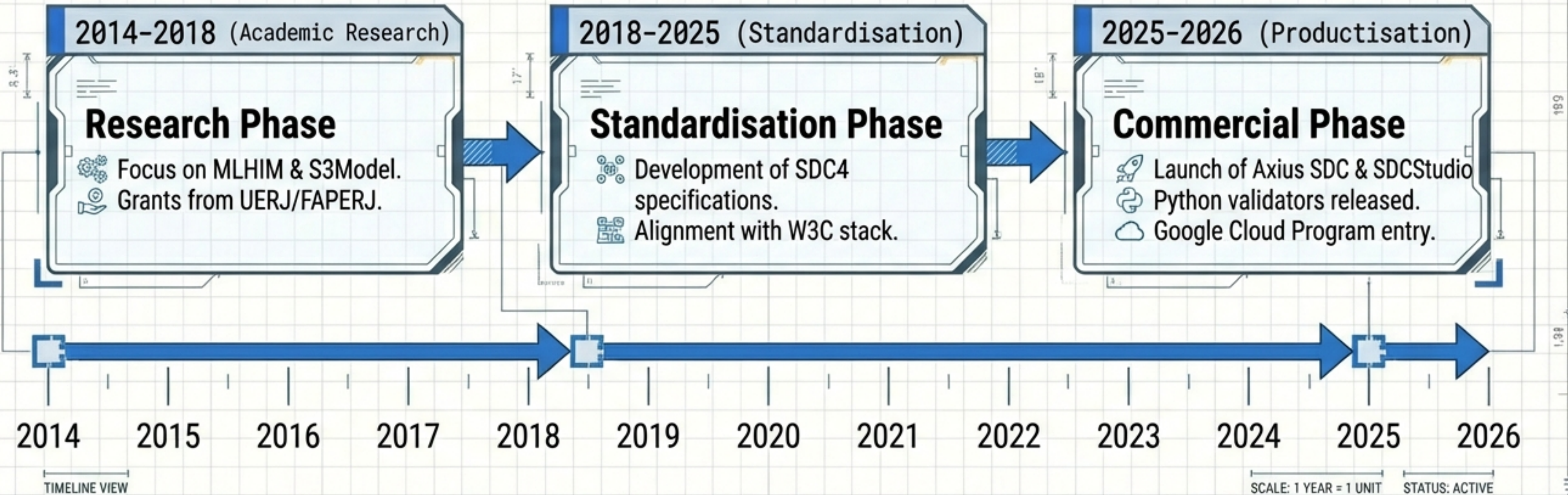
Planned Q2 2026



Knowledge Graph (`sdc-xml2graph`)

Planning Stage

The Evolution: Research to Commercialisation



⚠ Key Insight

Takeaway

Key Insight: While the SDCStudio software is new, the theoretical model has a decade-plus lineage in healthcare informatics.

Strategic Appraisal (SWOT Analysis)

STRENGTHS



Rigorous separation of concerns
(Structure vs. Semantics).

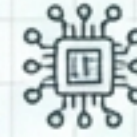
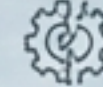


Built on immutable W3C standards
(XSD 1.1, RDF, SHACL).



'Schema-first' governance prevents drift.

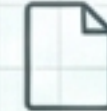
WEAKNESSES



High learning curve (XSD 1.1 is complex).



Dependency on single-vendor tooling
(SDCStudio).



Lack of verified third-party case studies.

OPPORTUNITIES



AI-driven data modelling (LLMs
generating strict schemas).



Legacy data migration markets
(Banking/Healthcare).



THREATS



Perception of 'Proprietary Standard'
hindering adoption.



Slow initial community uptake vs
JSON-schema.

References and Documentation

- ✂ **Specification:** [sdc4.xsd](#) (Normative Source of Truth)
- 🧠 **GitHub:** github.com/SemanticDataCharter
- 🧠 **Commercial:** axius-sdc.com (SDCStudio Documentation)

Disclaimer: This briefing is based on primary source analysis as of 29 January 2026.
Financial figures regarding R&D investment are self-reported by Axius SDC.

"The Semantic Data Charter represents a 'Governance-First' alternative to modern flexible schemas, trading initial speed for long-term architectural permanence."